

WEBSITE TRAFFIC ANALYSIS

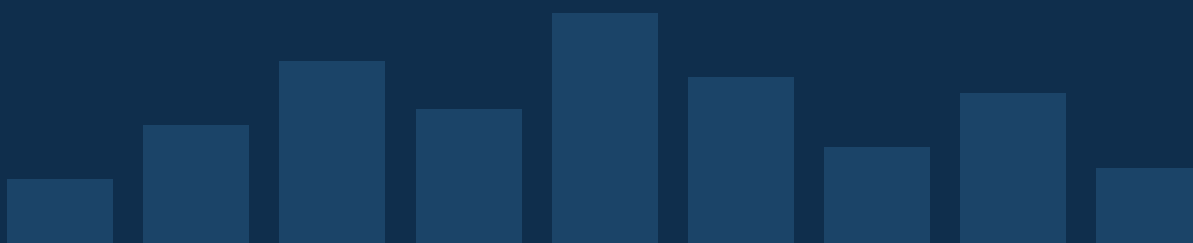
Alfido Tech – Data Analyst Internship

Prepared by

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TOOLS & TECHNOLOGIES

Python	Pandas	NumPy
Matplotlib	Seaborn	Google Colab



1. Project Overview

Website traffic data tells an organisation how people actually reach and move through its online content. This project analyses a real website traffic dataset to understand the volume of activity taking place, the type of actions visitors perform, where that traffic comes from geographically, and which content and links attract the most attention. The aim is to turn raw event logs into clear, usable insights that can support content and marketing decisions.

All findings in this report are taken directly from the analysis performed in the Google Colab notebook and are limited to the fields actually available in the dataset.

2. Objective

- Measure overall website traffic activity
- Understand the distribution of event types (pageview, click, preview)
- Identify daily traffic patterns across the period covered
- Examine the geographic distribution of traffic by country and city
- Find the most popular artists and tracks by traffic volume
- Identify the highest-performing links
- Describe user engagement patterns supported by the available dataset

3. Dataset

The project uses the **Website Traffic Analysis** dataset sourced from Kaggle. The raw file (`traffic.csv`) contains **226,278 rows and 9 columns**. After the cleaning steps described on the next page, the working dataset contains **122,567 rows and 9 columns**, covering the period **19 August 2021 to 25 August 2021**.

Column	Description
event	Type of interaction recorded (pageview, click, preview)
date	Date on which the event occurred
country	Country the traffic originated from
city	City the traffic originated from
artist	Artist associated with the content
album	Album associated with the content
track	Track associated with the content
isrc	Standard recording code for the track
linkid	Unique identifier of the link that generated the event

The dataset does not include user IDs, session IDs, referral sources, bounce rate or session duration. All analysis in this report is therefore based strictly on the nine fields listed above.

4. Tools & Technologies

Python was used as the core language, with **Pandas** and **NumPy** for data cleaning and aggregation, and **Matplotlib** and **Seaborn** for visualisation. The full analysis was carried out in **Google Colab**.

5. Data Cleaning

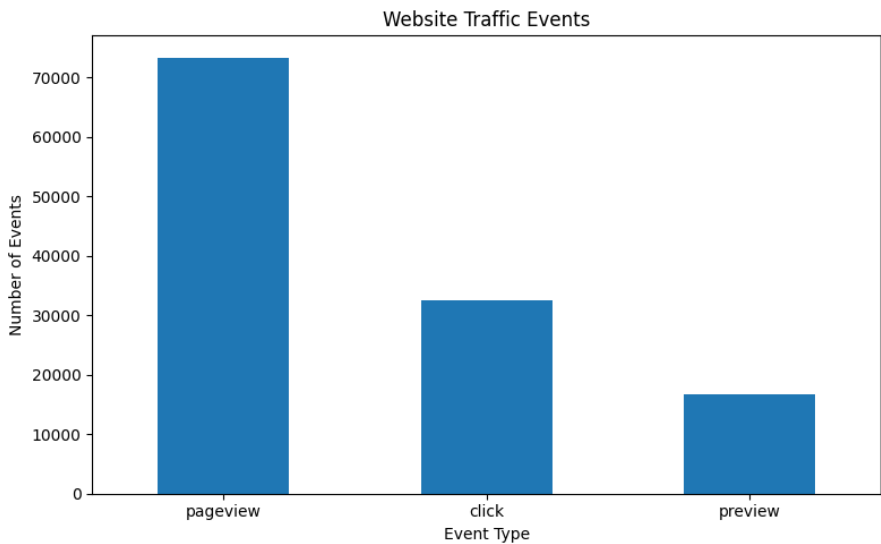
The following steps were performed in the notebook before analysis:

- Loaded the CSV dataset into a Pandas DataFrame (226,278 rows × 9 columns).
- Checked the dataset structure and previewed the first records using `info()` and `head()`.
- Checked missing values: country (11), city (11), artist (37), album (5), track (5) and isrc (7,121).
- Filled these missing values with the label “Unknown” so that no records were lost.
- Checked duplicate records, which accounted for 58.27% of the rows.
- Removed duplicate rows, reducing the dataset to 122,567 rows × 9 columns with 0 remaining duplicates and 0 missing values.
- Converted the date column into datetime format, confirming a range of 19–25 August 2021.
- Performed exploratory analysis and exported the cleaned dataset for reference.

6. Key Traffic Metrics

Total Traffic Events	122,567	Countries	212
Pageviews	73,360	Cities	11,993
Clicks	32,499	Artists	2,420
Previews	16,708	Tracks	3,563
		Links	3,839

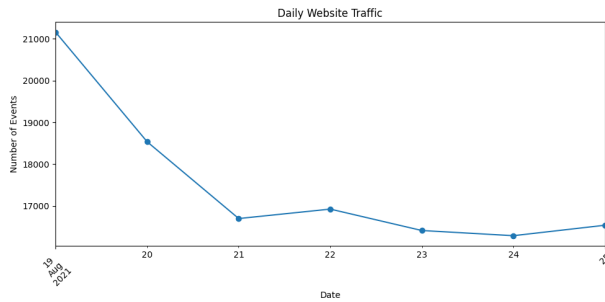
7. Event Distribution



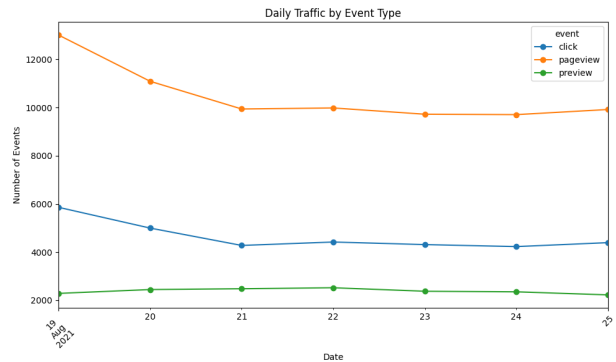
Pageviews are by far the most common event type with 73,360 events (59.85%), followed by 32,499 clicks (26.52%) and 16,708 previews (13.63%). This shows that most visitors view content, while a smaller share go on to click or preview.

8. Traffic & Content Insights

Daily Website Traffic

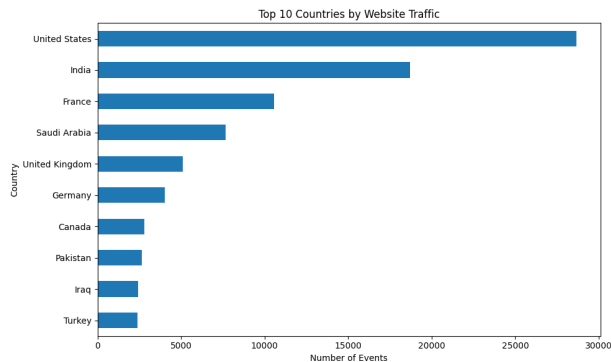


Traffic peaked on 19 August 2021 with 21,156 events, then fell to 18,536 on 20 August and settled between 16,291 and 16,927 events per day for the rest of the week.

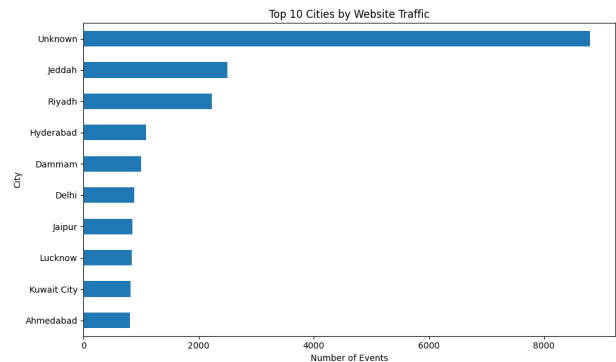


The same pattern holds for each event type. Pageviews dropped from 13,006 to around 9,700–9,900 per day, while previews stayed steady at roughly 2,200–2,500 events daily.

Geographic Distribution

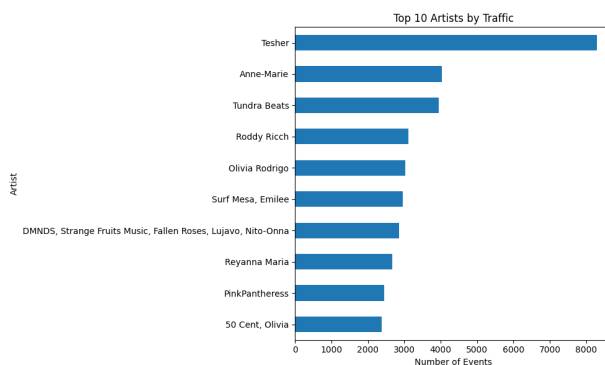


The United States generated the highest traffic with 28,664 events, ahead of India (18,689) and France (10,565). The top three countries account for a large share of all activity.

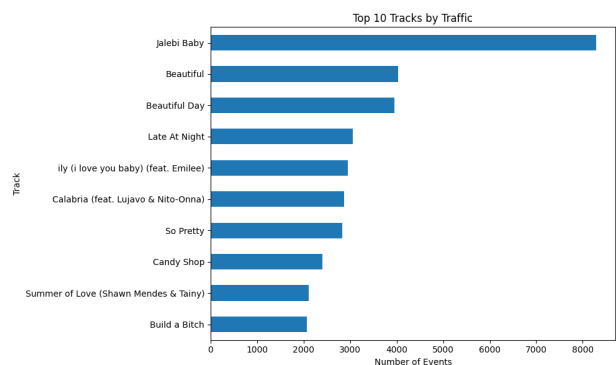


The largest city group is "Unknown" (8,802 events), where city data was not recorded. Among identified cities, Jeddah leads with 2,497 events, followed by Riyadh (2,232).

Popular Content



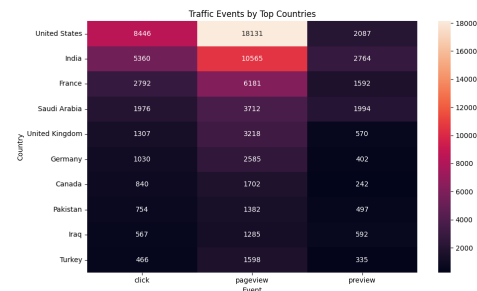
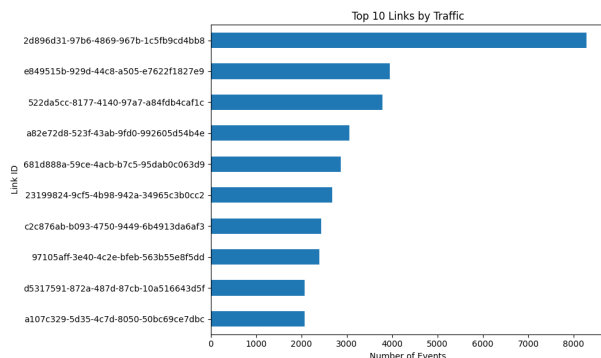
Teshar is the most interacted-with artist with 8,288 events, roughly double the next artist, Anne-Marie (4,029), followed by Tundra Beats (3,951).



"Jalebi Baby" is the most popular track with 8,288 events, well ahead of "Beautiful" (4,037) and "Beautiful Day" (3,951).

9. Link Performance

The dataset contains 3,839 unique links, but traffic is heavily concentrated: the single top-performing link recorded 8,288 events, more than twice the second link (3,950). A small group of links is responsible for most of the traffic reaching the site.



Left: the top 10 links by traffic. Right: events by type for the top 10 countries – the United States records 18,131 pageviews and 8,446 clicks, while India shows 10,565 pageviews and 5,360 clicks.

10. Key Findings

1. The cleaned dataset contains 122,567 traffic events recorded over seven days (19–25 August 2021), across 212 countries and 11,993 cities.
2. Pageviews dominate activity at 59.85% of all events, while clicks account for 26.52% and previews 13.63%.
3. Traffic was highest on the first day (21,156 events) and then stabilised at around 16,300–16,900 events per day.
4. The United States (28,664 events) and India (18,689 events) are the two largest traffic sources; Jeddah is the leading identified city.
5. Content and link traffic are highly concentrated: Tesher and the track “Jalebi Baby” both record 8,288 events, matching the single busiest link.

11. Recommendations for Alfido Tech

1. **Optimise the highest-traffic links and pages** – A small number of links drive most activity. Strengthening the calls-to-action on the top links offers the fastest route to more conversions.
2. **Improve click engagement** – Only about 1 in 4 events is a click. Clearer navigation and more interactive elements could help convert more of the 73,360 pageviews into actions.
3. **Run localised campaigns in high-traffic markets** – The United States, India, France and Saudi Arabia lead traffic volumes. Localised campaigns aimed at these markets, and at cities such as Jeddah and Riyadh, reach the largest existing audiences.
4. **Promote high-performing content** – Top content such as Tesher and “Jalebi Baby” attracts far more traffic than the rest. Featuring similar content in future campaigns builds on proven demand.
5. **Improve location tracking and monitor trends regularly** – City was unrecorded for 8,802 events, the largest city group. Better location capture and regular review of daily traffic trends would support more reliable decisions.

12. Conclusion

This analysis turned a large, duplicated raw traffic file into a clean dataset of 122,567 events and a clear picture of how the website is used – which event types dominate, how traffic behaves day to day, which countries and cities contribute most, and which artists, tracks and links perform best.

For Alfido Tech, this supports data-driven decision making: knowing where traffic comes from focuses marketing effort where the audience already is, knowing which content and links perform best shows what to promote and where conversion opportunities sit, and tracking the balance between pageviews, clicks and previews gives a simple, repeatable measure of engagement.